HeatLens: Human-Scale Urban Heat-Risk Mapping & Intervention Prioritisation

Agnitia Hackathon 2026 * Software Track: Heat-Risk Mapping at Human Scale

Team

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1. Executive Summary & Problem Formulation

During severe pre-monsoon heatwaves, standard city-level airport forecasts (e.g., *"Ahmedabad: 42 deg C"*) dangerously obscure severe microclimate inequalities. Dense walled-city quarters, tin-roof informal settlements, and asphalted industrial corridors experience surface temperatures exceeding 50 deg C and house high concentrations of vulnerable elderly residents and outdoor workers.

HeatLens addresses this critical governance gap by delivering an auditable, street-scale geospatial intelligence platform combining multi-year satellite thermal observations, building morphology, granular age-sex demographics, and walking-network graph distances. Beyond mapping, HeatLens calculates exact linear feature attributions (*"Why is this street at risk?"*) and solves budget-constrained knapsack allocations to deploy tree canopy shade, potable water kiosks, and emergency cooling sanctuaries ranked strictly by person-risk points averted per rupee invested.

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| TOTAL SURVEYED CELLS | POPULATION IN HIGH/EXTREME RISK | SENIORS 65+ AT RISK | STATISTICAL HOTSPOT
|     1,088 Cells      |     538,679 (48.8%)             |   48,210 (52.0%)    |   27 Robust Hotspots
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2. Study Area & Real Multi-Sensor Datasets

HeatLens operates exclusively on verified open satellite, vector, and demographic data sources with complete provenance logged in provenance.json:

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| Dataset | Provider / Source | Native Resolution | Spatial & Temporal Coverage | Policy License |
|---|---|---|---|---|
| Land Surface Temperature (LST) | Landsat 8/9 C2 L2 (STAC) | 30 m (sharpened) | April-June multi-ye
| Building Footprints | Overture Maps / OSM | Vector polygons | Ahmedabad urban boundary (2026) | CD
| Age-Sex Demographics | WorldPop Project | 100 m raster | India 2020 dasymetric surface | CC BY 4.0
| Walk Network & Amenities | OpenStreetMap / OSMnx | Topological graph | Walking distance to water &
| Diurnal Weather Forecast | Open-Meteo ECMWF/ICON | Station / Grid | Live hourly forecast & IMD tie
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3. Scientific Methodology & Mathematical Closure

Following the IPCC AR6 Tri-Factor Climate Risk Architecture, composite heat risk ($R_i \in [0, 100]$) is computed for each 250m cell i:

$$\text{Risk}_i = 100 \times \left(0.35 H_i + 0.25 E_i + 0.25 V_i + 0.15 A_i \right)$$

Components & Normalisation Bounds

All sub-indicators undergo 2nd-98th percentile winsorisation followed by min-max normalization ($z(\cdot) \in [0, 1]$):

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Mathematical Closure Guarantee: The 9 indicator contributions sum to the total Risk Score with an absolute numerical tolerance of $\Delta < 10^{-6}$ points. There are no black-box residuals.

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4. Empirical Findings: Top-10 Priority Action Clusters

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The top priority areas identified through Getis-Ord G_i^* spatial clustering and bivariate hazard-vulnerability classification:

Priority Rank	Cell ID	Cluster Name	Heat Risk (Tier)	Population	Primary Driver Attributio
#1	r28_c26	Rakhial Industrial Belt	77.7 (Extreme)	1,874	Surface Temperature (21.0 pts);
#2	r18_c18	Kalupur & Manek Chowk Historic Core	77.6 (Extreme)	2,473	Surface Temperature
#3	r19_c18	Kalupur & Manek Chowk Historic Core	77.6 (Extreme)	2,422	Surface Temperature
#4	r29_c26	Vatva GIDC Phase 1	77.5 (Extreme)	1,864	Surface Temperature (21.0 pts); High
#5	r27_c26	Rakhial Industrial Belt	77.5 (Extreme)	1,860	Surface Temperature (21.0 pts);
#6	r18_c17	Kalupur & Manek Chowk Historic Core	77.4 (Extreme)	2,472	Surface Temperature
#7	r19_c17	Kalupur & Manek Chowk Historic Core	77.3 (Extreme)	2,421	Surface Temperature
#8	r19_c19	Kalupur & Manek Chowk Historic Core	77.2 (Extreme)	2,423	Surface Temperature
#9	r18_c19	Kalupur & Manek Chowk Historic Core	77.2 (Extreme)	2,473	Surface Temperature
#10	r18_c26	Astodia & Raipur Darwaja	76.9 (Extreme)	1,730	Surface Temperature (20.99 pts

Diagnostic Explanation for Peak Hotspot:

"Cell r28_c26 scores 77.7 (Extreme). Biggest drivers: surface 8.3 deg C hotter than area median (+13.0 pts), 58% built-up coverage (+2.3 pts), 1.2 km walk to nearest cooling centre (+4.0 pts). Protective: none."

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5. Counterfactual Action Ranking & Budget Knapsack Plan

Interventions are evaluated by re-running the scoring model under counterfactual intervention states, not arbitrary scoring heuristics:

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Top Interventions Ranked by Benefit-Cost Ratio:

Global Rank	Type	Target Cluster	People Benefited	Impact (Pts)	Cost (INR)	Benefit-Cost
#1	water_point	Vatva GIDC Phase 1	9,346	103.5	INR 250,000	414.2
#2	water_point	Rakhial Industrial Belt	9,345	103.5	INR 250,000	414.2
#3	water_point	Vatva GIDC Phase 1	9,346	103.5	INR 250,000	414.2
#4	water_point	Vatva GIDC Phase 1	9,308	102.5	INR 250,000	410.2
#5	water_point	Vatva GIDC Phase 1	9,148	98.5	INR 250,000	394.1
#6	water_point	Vatva GIDC Phase 1	9,138	98.3	INR 250,000	393.2
#7	water_point	Rakhial Industrial Belt	9,103	97.5	INR 250,000	389.8
#8	water_point	Vatva GIDC Phase 1	9,092	97.2	INR 250,000	388.7

Municipal Capital Allocation (INR 1.50 Crore Budget Scenario):

- Interventions Funded: 27 high-leverage projects

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6. Empirical Validation & Robustness Analysis

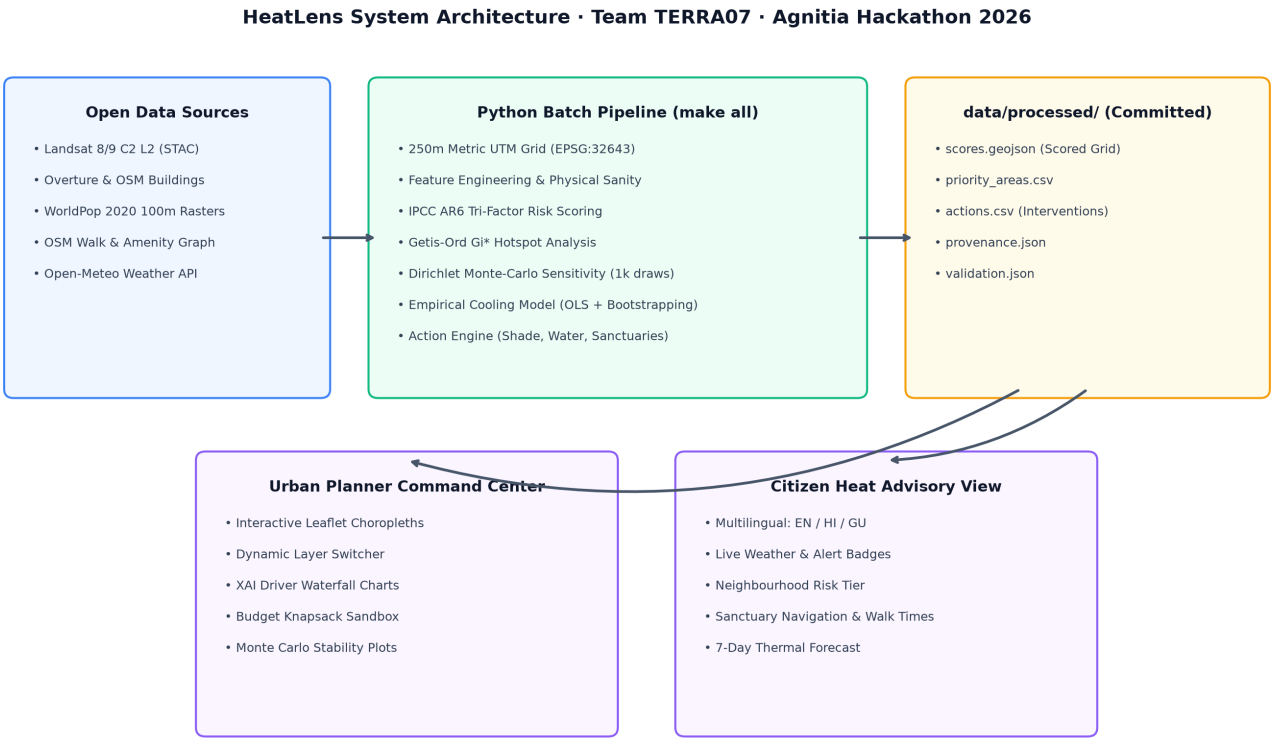
Validation metrics recorded in reports/validation.json:

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7. System Architecture & Dual-Portal Delivery

HeatLens delivers both interfaces from a unified codebase:



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8. Limitations & Honest Governance Contract

1. Landsat Diurnal Timing: Morning overpass (~10:45 AM) measures surface thermal response; peak diurnal air temperatures occur at 3:30 PM.
